Pergamon

Marine Pollution Bulletin, Vol. 36, No. 9, pp. 669–676, 1998
© 1998 Elsevier Science Ltd. All rights reserved
Printed in Great Britain

PII: S0025-326X(98)00058-7

0025-326X/98 $19.00+0.00

The Fine Sand *Abra alba* Community of the Bay of Morlaix Twenty Years after the Amoco Cadiz Oil Spill

J-C. DAUVIN

Station Marine de Wimereux, CNRS-UPRES-A ELICO, B.P. 80, 62930 Wimereux, France

The fine sand *Abra alba* community from the Bay of Morlaix (western English Channel) was strongly affected by the Amoco Cadiz oil spill of April 1978. The long term changes in the community (1977–1996) show that reconstitution of this community is slow (over 10 yr). A progressive recolonization by amphipod *Ampelisca* populations constituting the dominant species is observed. The results show that it is necessary to survey the affected communities for a long period of time (> 10 yr) after an event to identify the real ecological impact of an oil spill. The abiotic and biotic factors existing in the Bay of Morlaix are favorable to the establishment and the persistence of stable dominant *Ampelisca* populations in time. The 'climax' concept seems to be applicable to this type of stable environment. The soft-bottom *Abra alba* community of the Bay of Morlaix may only support a maximal carrying capacity of approximately 10–11 g m^{-2} (mean annual biomass in decalcified dry weight). The deficit of production during 11 annual cycles may have affected the fish biomass feeding on this community. **© 1998 Elsevier Science Ltd. All rights reserved**

Keywords: *Abra alba* community; Amoco Cadiz; oil spill; long term survey; secondary production; English Channel

In the Bay of Morlaix, the Pierre Noire fine sand community supports the richest benthic amphipod community yet recorded in European seas, with very dense populations of *Ampelisca* (Dauvin *et al.*, 1993). In 1977, a benthic survey of temporal changes of this community was initiated. In spring 1978, 1 yr after the beginning of sampling, the site was polluted by hydrocarbons from the Amoco Cadiz, resulting in the disappearance of the dominant *Ampelisca* populations. A single species remaining just after this event was found at very low densities (Dauvin, 1987). In the English Channel, a megatidal sea, fine sediments are confined to the shallow waters of bays and estuaries. Fine-sand and muddy fine sand communities such as the fine-sand community of Pierre Noire are therefore isolated communities, separated by other substrata (rocky, pebbly and gravely, coarse sand, marl). Amphipods, characteristic of these communities, lack pelagic larvae, and therefore maintain insular populations (Dauvin, 1987). A 20 yr survey (1977–1996) at the Pierre Noire station with regular sampling of temporal changes of the community is now available. The objectives of this paper on this dataset are (i) to describe changes in total abundance and biomass, (ii) to estimate the macrobenthic production, and (iii) to estimate the impact of the deficit of benthos production on the fish biomass feeding on the Pierre Noire community.

Materials and Methods

Study area

The Pierre Noire (PN) site, in the *Abra alba–Hyalinoecia bilineata* fine sand macrobenthic community, is located in the eastern part of the Bay of Morlaix, Brittany (western English Channel) (48°42, 50′N; 3°51, 96′W), 17 m depth at low tide. The sediment is fine sand and shows low temporal fluctuations (median particle size: 148–184 μm; percentage of particles <63 μm $<1\%$). In the western part of the English Channel, along the French coast, the Atlantic water influence and strong tidal currents produce a homogeneous water column along the year with an absence of summer thermocline. Hydrological characteristics present few seasonal and pluri-annual fluctuations (Dauvin, 1984). The bottom water temperature varied usually between 8°C in March to 15°C in September, and the salinity between 34.5 PSU in winter and 35.3 PSU at the beginning of October (Dauvin, 1984).

Sampling

Ten replicate samples (Smith McIntyre grab, 0.10 m^2) were taken at the PN station five times per year at the beginning of March, June, August, October and December to estimate the abundance and biomass of the macrofauna from 1977 to 1996 (20 yr). After collection, the sediment was sieved (1 mm circular mesh) and the retained material was fixed with 10%

neutralized formalin. A total of 1000 Smith McIntyre samples at 100 dates are thus available.

Laboratory analysis

The retained material was sorted twice, and identified to species level. The mean annual abundance (individuals m^{-2}), and the mean annual biomass in $g\,m^{-2}$ decalcified dry weight (DDW; Dauvin, 1984) were estimated for each of the 20 annual cycles (mean annual values from March to December sampling). The annual production was estimated using the mean annual biomass and the P/B values of the dominant species or main zoological groups. These values were identified from Bay of Morlaix species or values found in the literature (Table 1, Dauvin, 1984, 1991). The species were grouped into four trophic groups.

1. M, mixed macrobenthic invertebrates (containing suspension feeders, deposit feeders and suspension/deposit feeders following the suggestions of Arntz and Brunswig, 1975). They feed on organic matter, originating mainly from plankton production, found at the sediment surface and to a lesser degree, on microphytobenthos and benthic bacteria.
2. DSS, the subsurface-deposit feeders feeding on detritus and bacteria in the sediment.
3. C, carnivores, motile predators.
4. O, omnivores and necrophages feeders feeding on carrion (Chardy and Dauvin, 1992).

Demersal fish biomass and consumption

Stomach content analysis of the 13 main species of demersal fish showed that they feed on macrobenthic fauna, especially amphipods, polychaetes and bivalves (Dauvin, 1988). On the Pierre Noire fine sand community, it was estimated that about 50% of the benthic biomass was available for demersal fish consumption (Dauvin, 1988). Dry weight was estimated to be 20% of fresh weight. Daily consumption of macrobenthos by fish was estimated at 3% of the fish biomass (Arntz, 1980).

Oil spill

Of the 220000 t of oil spilled into the sea from the Amoco Cadiz wreck, between 10000 and 92000 t, according to different authors, became trapped in subtidal sediments (Dauvin, 1984). The presence of hydrocarbons in the subtidal bottom sediments of Morlaix Bay was evident at the beginning of April 1978, just 2 weeks after the wreck (Cabioch *et al.*, 1978). At Pierre Noire, the levels of hydrocarbons (in part 10^{-6}, i.e. $mg\,kg^{-1}$ dry sediment) in the sediments measured by infrared spectrophotometry reached 200 ppm in the summers of 1978 and 1979, and did not exceed 50 ppm after the winter of 1981 (Dauvin, 1984).

Results

Abundance

Fig. 1a shows the temporal changes of the mean annual abundance of the Pierre Noire community from 1977 to 1996. The relative dominance of *Ampelisca* in 1977 and from 1990 to 1996 was approximately 75% of the total abundance, and reached in 1977, 1993 and 1995, 90% of the total abundance. Polychaetes of the families Spionidae and Paraonidae succeeded in dominance. However, two species of Spionidae,

TABLE 1

Production/biomass (P/B) ratio and trophic groups for the main species or zoological groups estimated at the Pierre Noire Station (*) or reported in literature (see Dauvin, 1991). M, mixed feeders; DSS, subsurface deposit feeders; C, carnivorous species; O, omnivorous species.

Species	P/B	Group	Species	P/B	Group
Polychaete			Molluscs		
Ampharete acutifrons	3.6	M	*Abra alba**	2.1	M
Aricidea cerrutii	3.0	M	*Abra prismatica**	2.8	M
Aricidea fragilis	3.0	M	*Cultellus pellucidus*	2.8	M
Chaetozone setosa	1.3	M	*Thyasira flexuosa*	1.4	M
Euclymene oerstedii	1.2	DSS	*Venus ovata**	1.6	M
Eulalia sanguinea	3.0	O	Bivalves	2.0	M
Heterocirrus alatus	4.5	DSS	*Hinia reticulata*	1.5	O
Hyalinoecia bilineata	1.2	C	Gastropods	1.2	O
Lanice conchilega	1.2	M	Crustacean		
Marphysa bellii	1.2	C	*Ampelisca armoricana**	2.25	M
Nephtys hombergii	1.5	C	*Ampelisca brevicornis**	2.6	M
Notomastus latericeus	3.0	DSS	*Ampelisca sarsi**	2.4	M
Paradoneis armata	3.0	M	*Ampelisca tenuicornis**	3.7	M
Phyllodoce groenlandica	1.2	C	*Ampelisca typica**	4.2	M
Polydora pulchra	4.9	M	*Perioculodes longimanus*	3.0	M
Scoloplos armiger	3.0	DSS	*Urothoe pulchella*	3.0	M
Spio decoratus	3.0	M	Amphipods	3.0	M
Siophanes bombyx	2.5	M	*Pagurus bernhardus*	1.2	O
Errantia polychaete	1.2	C	Peracarids	3.0	M
Sedentaria polychaete	3.0	DSS	Decapods	1.2	O
Miscellaneous	1.0	C	Echinoderms	1.2	M
			Ophiura albida	2.8	O

Polydora pulchra and *Spio decoratus*, showed important abundances. In 1982, a summer peak in abundance of the opportunistic polychaete *Polydora pulchra* was observed, thus the annual abundance of Spionidae exceeded 8000 ind. m^{-2}. The total abundance of Spionidae exceeded 1000 ind. m^{-2} in 1982, from 1984 to 1987, and from 1990 to 1994. The mean annual abundance of Paraonidae remained lower than 600 ind. m^{-2} except in 1978 when it reached 650 ind. m^{-2}. After this, values decreased regularly until the end of the observations in 1996 when annual abundance was lower than 50 ind. m^{-2}. The total abundance exceeded 24000 ind. m^{-2} in 1977, decreasing rapidly in 1978 just after the Amoco Cadiz oil spill due to the disappearance of the *Ampelisca* population. The mean annual total abundance showed very low values (<5000 ind. m^{-2}) until 1985 with a minimum of 2938 ind. m^{-2} in 1979. Nevertheless, an exceptional value of 11903 ind. m^{-2} was reached in 1982 as a result of the peak of *Polydora*. Values remained lower than 10000 ind. m^{-2} between 1986 to 1989. Recolonization by *Ampelisca* amphipods took place after 1989, dominating total abundance of the community. The annual abundance remained higher than 12500 ind. m^{-2} from 1990 to 1996. In 1993, it reached 28000 ind. m^{-2} in the same order of magnitude as in 1977 (> 24000 ind. m^{-2}).

Although the species number was very high, with more than 420 species reported in the community during the 20 yr survey, only 25 species were among the first ten dominant species at each annual cycle (Table 2). Only six species, *Ampelisca sarsi* (six times), *Spio decoratus* (five times), *A. armoricana* and *A. tenuicornis* (three times each), *Polydora pulchra* (twice) and *Paradoneis armata* (once) occupied successively the first rank. *Ampelisca* species occupied the three first ranks before the pollution event of 1977, then again at the end of the survey from 1991 to 1996. There was a successive dominance of *Ampelisca*: *A. armoricana* was dominant in 1977 then again in 1987–1988, *A. sarsi* in 1981 and then again from 1989 to 1993, after which *A. tenuicornis* succeeded at the first rank for the last 3 yr. In 1978, the polychaete, *Paradoneis armata* replaced *Ampelisca*, then both Spionidae *Spio decoratus* and *Polydora pulchra* successively occupied the first rank. One species, *Spio decoratus* was always in the first ten

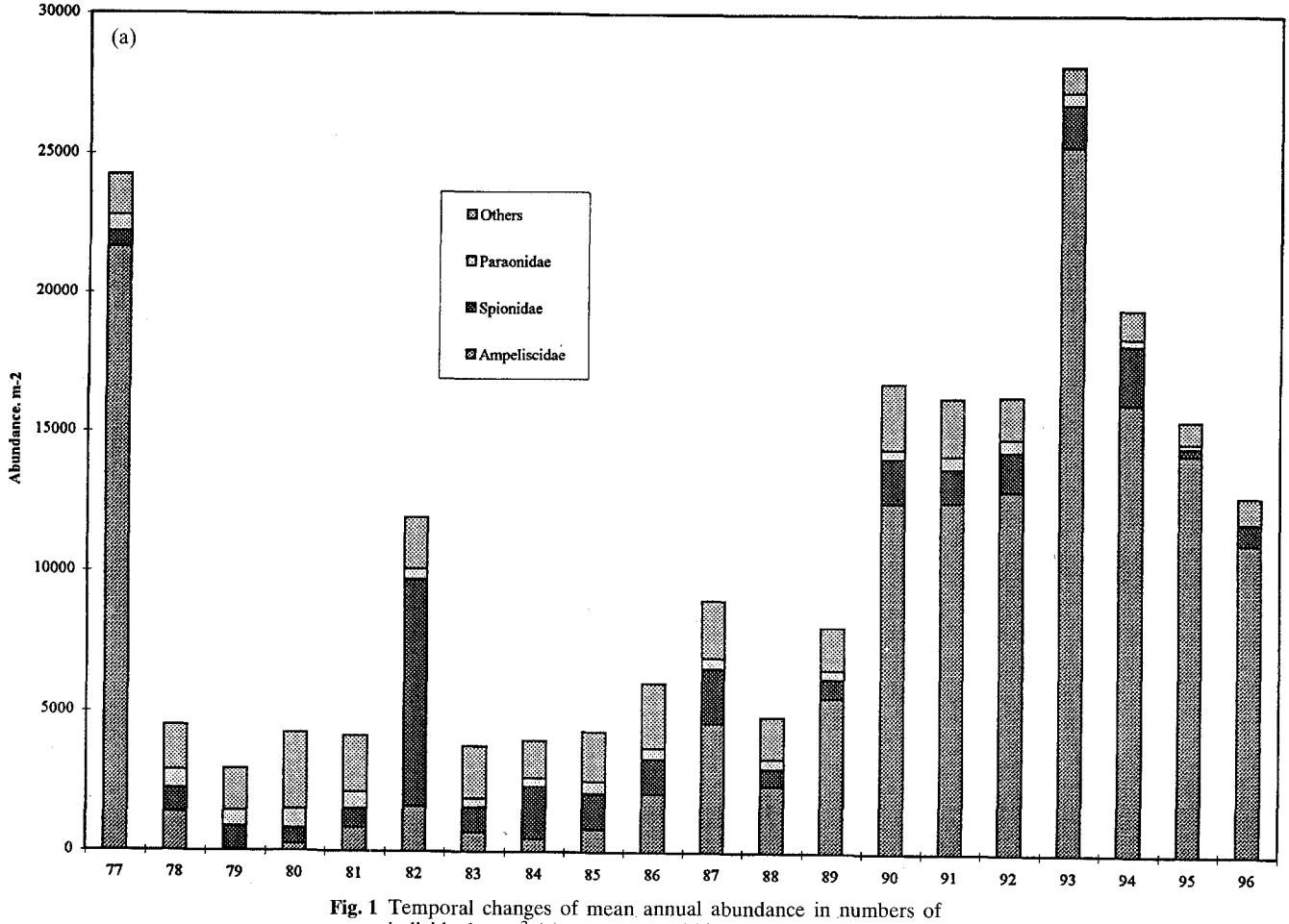


Fig. 1 Temporal changes of mean annual abundance in numbers of individuals m^{-2} (a), mean annual biomass in g decalcified dry weight m^{-2} (b), and annual production in g $m^{-2} yr^{-1}$ (c) of the fine sand Pierre Noire community from 1977 to 1996. (O: omnivores; C: carnivores; DSS: subsurface deposit feeders; M: mixed)

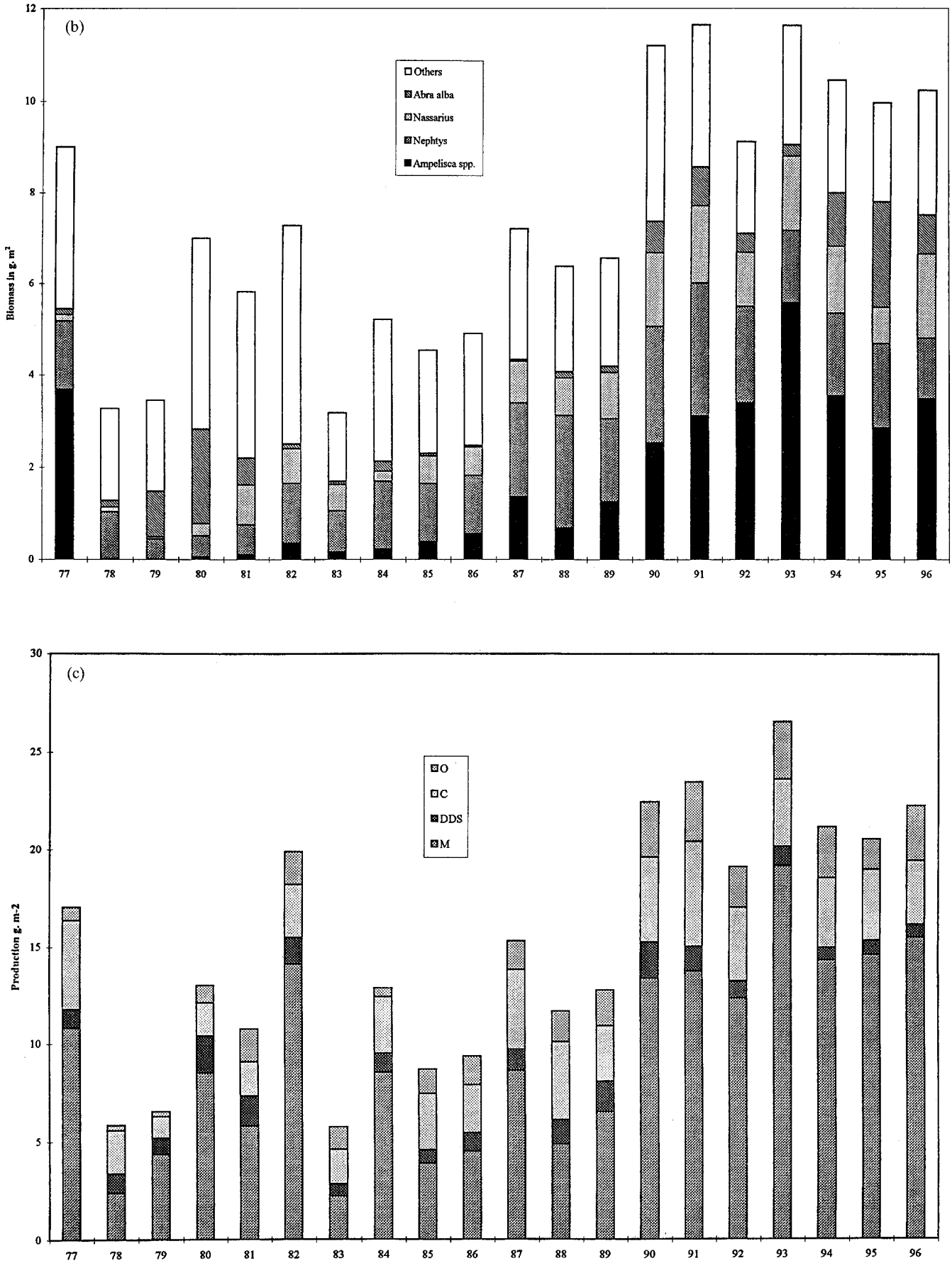


Fig. 1 Continued

TABLE 2

Abundance rank and total dominance of the first ten species for each annual cycle.

Species	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96
Spio decoratus	4	2	1	1	4	7	1	3	1	1	4	3	3	3	4	4	4	5	4	4
Paradoneis armata	5	1	2	2	2	4	4	6	5	6	7	6	5	10	6	5	7	7	6	—
Ampelisca sarsi	2	—	—	7	1	2	2	2	2	3	2	2	1	1	1	1	1	2	2	2
Ampelisca armoricana	1	—	—	—	—	—	—	—	7	2	1	1	2	2	2	2	3	3	3	3
Ampelisca tenuicornis	3	—	—	—	—	—	—	—	—	—	—	—	—	5	3	3	2	1	1	1
Polydora pulchra	—	8	—	—	3	1	5	1	3	—	3	4	8	4	5	6	5	4	—	5
Marphysa bellii	6	6	6	4	7	10	—	—	—	8	—	8	10	—	—	—	—	—	9	8
Aricidea fragilis	7	5	7	10	9	—	8	10	10	—	10	9	—	—	—	9	8	10	—	—
Chaetozone setosa	9	3	3	3	5	—	7	—	—	10	—	7	4	6	8	8	10	—	8	9
Ampelisca brevicornis	10	—	—	—	—	8	10	8	—	7	6	—	—	—	—	—	—	—	—	—
Urothoe pulchella	—	—	—	9	6	3	3	5	4	5	8	—	—	—	—	—	—	—	—	—
Scoloplos armiger	—	10	5	6	8	—	6	9	—	—	—	—	—	—	—	—	—	—	—	—
Ampelisca typica	—	—	—	—	10	5	—	—	6	4	5	—	9	—	—	7	6	6	10	6
Nephtys hombergii	—	—	—	—	—	6	—	4	8	9	9	5	7	9	—	—	—	8	7	10
Abra alba	—	—	4	5	—	—	—	—	—	—	—	—	—	8	7	—	—	9	5	7
Heterocirrus alatus	—	4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Thyasira flexuosa	—	—	8	8	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Spiophanes bombyx	—	—	—	—	—	—	9	7	—	—	—	10	6	—	10	10	—	—	—	—
Aricidea cerrutii	—	7	10	—	—	—	—	—	—	—	—	—	—	—	—	—	9	—	—	—
Ampharete acutifrons	8	—	—	—	—	—	—	—	—	—	—	—	—	—	9	—	—	—	—	—
Perioculodes longimanus	—	9	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Abra prismatica	—	—	9	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Lanice conchilega	—	—	—	—	—	9	—	—	—	—	—	—	—	—	—	—	—	—	—	
Bathyporeia elegans	—	—	—	—	—	—	—	—	9	—	—	—	—	—	—	—	—	—	—	—
Polydora flava	—	—	—	—	—	—	—	—	—	—	—	—	7	—	—	—	—	—	—	—
Total dominance	95	80	79	71	70	90	72	77	68	71	80	77	84	87	88	90	97	96	96	95

species. Four species occurred in the top ten species between 15 and 19 times, and six species appeared only once (see Table 2).

Few species dominated in the Pierre Noire community. During the first annual cycles after the event the polychaetes of the families Paraonidae and Spionidae, in 1977 just behind *Ampelisca,* became the dominant species. Disappearance of the dominant species due to the oil pollution formed a large vacant ecological niche. One opportunistic polychaete species *Heterocirrus alatus* showed a temporal peak of abundance in autumn 1978 (Dauvin, 1984). The temporal changes showed a progressive recolonization of the station by *Ampelisca. Ampelisca sarsi* remained low in number just after the event, being the first to reconstitute its population, probably due to a favourable partitioning of resources (Dauvin *et al.*, 1993). *Ampelisca tenuicornis*, the latest to recolonize the community became the dominant species of the community, as the favourable bivoltinism allowed a rapid population revival.

Biomass

Figure 1b displays the temporal changes of the mean annual biomass of the Pierre Noire community from 1977 to 1996 showing the dominant taxa: the *Ampelisca* amphipods, the polychaete *Nephtys hombergii,* the gastropod *Hinia reticulata,* the bivalve *Abra alba,* and the remaining species. In 1978, there was a reduction of about 70% of the biomass mainly due to the disappearance of *Ampelisca* spp. The biomass remained low in 1979, then doubled in 1980 as a result of an increase in *Abra alba* abundance. From 1981 to 1990, the mean annual biomass fluctuated between 7.3 g m^{-2} in 1982 (high contribution of *Polydora pulchra*) and a very low value in 1983, 3.2 g m^{-2}, which was the lowest value of the 20 yr survey. The biomass of *Ampelisca* increased significantly until 1989 and reached a maximum of 5.6 g m^{-2} in 1993 and fluctuated from 2.5 to 3.5 g m^{-2} over the last 3 yr. The mean annual biomass was relatively stable from 1990 to 1996, fluctuating between 9.2 g m^{-2} in 1992 and 11.73 g m^{-2} in 1991 and 1993. The overall mean annual biomass during the last 7 yr of the survey was 10.65 g m^{-2}.

Nephtys hombergii biomass showed low values from 1979 to 1981, then increased more or less regularly from 1982 to 1991 when the biomass reached 2.9 g m^{-2} compared with 0.45 g m^{-2} in 1979 and 1980. The contribution of this carnivorous species decreased just after 1991 until 1996 reaching 1.3 g m^{-2} in 1996. There was also a general trend increasing with time, of the necrophagous species *Hinia reticulata* from low values at the beginning of the observations in 1977–1980 (<0.3 g m^{-2}) to values <1 g m^{-2} from 1981 to 1988. Values >1 g m^{-2} were observed at the end of the survey between 1989 and 1996 except in 1995 when the biomass was 0.8 g m^{-2}. The biomass of *Abra alba* fluctuated more dramatically from a minimum of 0.03 g m^{-2} in 1986 to a maximum of >2.0 g m^{-2} in 1980 and 1995. Biomass related to the rest of the species fluctuated from a minimum of 1.5 g m^{-2} in 1983 to a maximum of 4.8 g m^{-2} in 1982 (peak of *Polydora*). The biomass of this group was found 12

times to be between 2 and 3 g m^{-2}, seven times between 3 and 4 g m^{-2} and only twice > 4 g m^{-2}.

Twenty-four species were among the ten first dominant species in biomass during the survey (Table 3). Only six species occupied the first rank: *N. hombergii* (12 times), *A. alba* (3), *Ampelisca armoricana* (2), *Ampelisca tenuicornis*, *Hinia reticulata* and *Polydora pulchra* (1). Two species *N. hombergii* and the polychaete *Marphysa bellii* were always among the dominant species. *H. reticulata* was present 19 times, seven other species were among the ten first species at least ten times. Three species were present in the dominant species only once, and four others only twice. There was a succession of dominant species following a decrease in abundance of the sipunculid *Phascolion strombi*, the temporal dominance of three polychaetes *Scoloplos armiger*, *Paradoneis armata* and *Chaetozone setosa* during the first years after the pollution event, and the recolonization of *Ampelisca* at the end of the survey with *A. tenuicornis* which became the dominant species in 1996.

Few species dominated in biomass in the Pierre Noire community. There was a rapid decrease of biomass in 1978 due to the disappearance of *Ampelisca* spp., then during the first annual cycles after the event, the bivalve *A. alba* and polychaetes increased in biomass. However, the carnivorous polychaetes *N. hombergii*, *M. bellii* and *Hyalinoecia bilineata*, and the necrophagous species *Hinia reticulata* always occupied the first ranks in biomass of the community. *Ampelisca armoricana*, dominating in 1977, was replaced as the first species in terms of abundance and biomass in 1996 by *A. tenuicornis* once stable conditions had returned.

Production

Figure 1c shows temporal changes in annual production of the Pierre Noire community with the corresponding four trophic groups: mixed, subsurface deposit-feeders, carnivores and omnivores. Fluctuations in production were in accordance with changes in biomass with a dramatic decrease from 1977 (17 g m^{-2} yr^{-1}) to 1978 (5.8 g m^{-2} yr^{-1}), low values (< 15 g m^{-2} yr^{-1}) from 1979 to 1989 except in 1982 (19.9 g m^{-2} yr^{-1}) and 1987 (15.3 g m^{-2} yr^{-1}) with high production of *Polydora pulchra*, and values > 20.0 g m^{-2} yr^{-1} during the last 7 yr of the survey. The annual production reached its maximum in 1993 with 26.6 g m^{-2} yr^{-1}. Two trophic groups dominated biomass and production: M and C which formed 62% and 21%, respectively, of the total production of the community. DSS was very low about (7% of production); omnivores were higher, representing 11%.

The P/B ratio fluctuated between 1.79 in 1978 and 2.73 in 1982. The overall mean was 2.03. The maximum occurred when the polychaete *Polydora pulchra* dominated the community and minimum when the mixed feeders were minimal between 1978 and 1981. The P/B ratio was more stable during the last 7 yr of the survey when *Ampelisca* dominated the community (at a value of ~2).

Fish biomass

Thirty-one species of fish were collected from 1977 to 1981 over 16 trawls, the fish community was dominated by 13 species of demersal fishes (Dauvin, 1988). The percentages of numerical dominance on 2432 individuals fished, were as follows: *Trisopterus*

TABLE 3

Biomass rank and total dominance of the first ten species for each annual cycle.

Species	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96
Nephtys hombergii	2	1	2	3	2	2	1	1	1	1	1	1	1	1	1	1	4	1	2	3
Hinia reticulata	7	6	—	7	1	3	2	7	2	2	3	2	2	2	2	4	3	2	5	2
Ampelisca armoricana	1	—	—	—	—	—	—	—	10	3	2	3	3	3	4	2	1	3	3	4
Ampelisca tenuicornis	5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	5	2	5	4	1
Abra alba	9	4	1	1	3	10	10	3	—	—	—	7	8	5	5	6	6	4	1	5
Polydora pulchra	—	—	—	—	—	1	8	2	3	—	4	—	—	7	8	—	9	7	—	9
Marphysa bellii	3	3	3	5	7	4	3	4	4	7	5	8	7	10	7	7	10	8	8	7
Lanice conchilega	—	—	4	2	5	5	—	5	—	—	9	4	6	6	9	—	—	10	9	—
Ampelisca sarsi	4	—	—	—	—	7	4	10	7	9	8	6	4	4	3	3	5	6	6	6
Phascolion stombi	6	2	5	6	4	6	5	—	—	—	—	—	10	8	—	—	—	—	—	—
Hyalinoecia bilineata	8	7	—	—	—	—	—	—	—	—	—	10	—	—	—	10	7	9	7	8
Spio decoratus	—	9	6	—	—	—	6	6	5	8	10	—	—	7	—	9	8	—	—	—
Ampelisca brevicornis	—	—	—	—	—	8	—	8	6	4	6	—	—	—	—	—	—	—	—	—
Scoloplos armiger	—	10	7	8	8	9	7	9	—	—	—	—	—	—	—	—	—	—	—	—
Euclymene oerstedii	—	—	—	—	10	—	—	—	9	5	7	5	5	9	8	8	—	—	10	—
Venus ovata	10	—	10	4	9	—	—	—	—	—	—	—	—	—	—	—	—	—	—	10
Paradoneis armata	—	5	9	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Chaetozone setosa	—	8	8	10	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Cultellus pellucidus	—	—	—	9	6	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Ampelisca typica	—	—	—	—	—	—	—	—	—	6	—	—	—	—	6	—	—	—	—	—
Urothoe pulchella	—	—	—	—	—	—	9	—	—	—	—	—	—	—	—	—	—	—	—	—
Ampharete acutifrons	—	—	—	—	—	—	—	—	8	—	—	—	—	—	—	—	—	—	—	—
Abra prismatica	—	—	—	—	—	—	—	—	—	10	—	—	—	—	—	—	—	—	—	—
Spiophanes bombyx	—	—	—	—	—	—	—	—	—	—	—	9	9	—	—	—	—	—	—	—
Total dominance	81	66	70	71	68	73	66	69	63	64	74	74	76	78	79	84	85	85	84	80

minutus (45%), *Callionymus lyra* (29.1), *Echiichthys vipera* (7.1), *Merlangius merlangus* (3.5), *Solea vulgaris* (3.2), *Pomatoschistus* spp. (2.3), *Pleuronectes platessa* (2.0), *Trisopterus luscus* (1.6), *Raja* spp. (1.2), *Mullus surmuletus* (1.0), *Aspitrigla cuculus* (0.6), *Scophthalmus rhombus* (0.6), *Scyliorhinus canicula* (0.4). These fish feed on macrobenthic fauna, especially crustacean amphipods, polychaetes and bivalves, and it has been estimated that about 50% of the benthic biomass is available for demersal fish (Dauvin, 1988). Table 4 gives the annual production, the annual available production for fish and the mean annual estimated fish biomass in accordance with a ratio of 10% between the available production and the fish biomass during the 20 yr span of the survey. The estimated fish biomass fluctuated between a minimum value of 0.13 g fresh weight m^{-2} in 1978 and 1983, and a maximum value of 0.61 g fresh weight m^{-2} in 1993. The mean annual fish biomass remained lower than 0.40 g fresh weight m^{-2} during the 12 yr after the pollution event except in 1982. The values of estimated fish biomass (Table 4) were in the same order of magnitude of the calculated fish biomass from the 1977–1981 trawls: 0.1–0.5 g fresh weight m^{-2} (Dauvin, 1988) with an annual fish consumption comprised between 1.1 and 5.5 g fresh weight m^{-2}. It is clear that just after the event the macrobenthic production could not support a fish biomass higher than 0.1–0.3 g fresh weight m^{-2}. In normal years such as before 1977 or during the last 7 yr of the survey from 1990 to 1996, the fresh weight fish biomass was greater than 0.4 g fresh weight m^{-2} and reached 0.6 g fresh weight m^{-2} in 1996. The deficit of macrobenthic production during 11 annual cycles after the pollution (1978–1989) ranged from 6 to 16 g m^{-2}, and deficit of fish biomass was calculated at values between 0.16 and 0.37 g fresh weight m^{-2} (mean 0.26 g fresh weight m^{-2}). The area covered by the Pierre Noire fine sand community is 2.5 km^2 (Dauvin, 1984) so the deficit of fish biomass of the community was estimated to 0.65 t fresh weight during the 11 yr period. Assuming a commercial value of about 10 FF (in 1998) per kilo, the deficit of the fishery can be estimated at 650×10 FF $\times 11$ annual cycles = 71500 FF ($\cong$US$ 12000) for the 11 yr period following the pollution event. Although this estimate appears low, it corresponds only to a limited area, particularly productive, of a larger zone of the north Brittany coast affected by the Amoco Cadiz oil spill.

Discussion

The Pierre Noire community was strongly polluted by the Amoco Cadiz in 1978. There were very evident reductions in abundance, biomass and production of the community by the disappearance of the dominant population of amphipods *Ampelisca* which are very sensitive to oil. The sediment was rapidly depolluted, and in 1981 benthic recruitment occurred in normal conditions. However, recolonization by *Ampelisca* was retarded due to the combination of different demographic strategies of amphipods (low dispersal of organisms, low fecundity), and the distance from non-perturbed populations which could supply recruits (Dauvin, 1987). Nevertheless, the populations showed a very high capacity to recolonize the affected site and recovered within 15 yr to achieve similar high densities to those found before the pollution incident. The duration of recolonization of affected populations after an oil spill seems in several cases to surpass 10 yr (Smith, 1968; Laubier, 1991; Holloway and Horgan,

TABLE 4

Biomass, production, P/B ratio, and available production for fish consumption assuming that approximately 50% of the benthic biomass was available to demersal fish over the 20 yr of the survey with a daily consumption of fish at 3% of the fish biomass according to Arntz (1980). Annual values in mg decalcified dry weight excluding annual fish consumption and fish biomass where values are in mg fresh weight.

Annual cycle	Biomass	Production	P/B	Edible production	Fish consumption	Fish biomass
1977	9012	17054	1.89	8527	4264	389
1978	3271	5853	1.79	2926	1463	134
1979	3448	6563	1.90	3282	1640	150
1980	7011	13019	1.86	6510	3255	297
1981	5824	10767	1.85	5384	2692	246
1982	7301	19899	2.73	9950	4975	454
1983	3187	5768	1.81	2884	1442	132
1984	5227	12900	2.47	6450	3225	295
1985	4555	8705	1.91	4352	2176	199
1986	4920	9365	1.90	4683	2341	214
1987	7245	15322	2.11	7661	3831	350
1988	6426	11678	1.82	5839	2919	266
1989	6606	12773	1.93	6387	3193	292
1990	11273	22480	1.99	11240	5620	513
1991	11728	23489	2.00	11745	5873	536
1992	9192	19152	2.08	9576	4788	437
1993	11726	26582	2.27	13291	6646	607
1994	10535	21205	2.01	10603	5301	484
1995	10054	20571	2.05	10286	5143	470
1996	10025	22301	2.22	11151	5575	509

1991). The implications of this delay are that ecological surveys after a pollution event must be continued for a sufficient time. This is not often compatible with the length of the financial grants. For the Amoco Cadiz, the research programmes had been financed from 1978 to 1981 by the NOAA/CNEXO programmes then from 1981 to 1984 by the 'Veille Ecologique' programme supported by the French Environmental Ministry. The Pierre Noire survey continued without support from 1984 to 1990. Later, support was restarted by French Environmental Ministry under the 'Réseau National d'Observation' co-ordinated by the IFREMER.

In the case of benthic communities, ampeliscids often live sympatrically, so when dense populations form, two or three species are simultaneously very abundant (Dauvin *et al.*, 1993). The fact that sympatric *Ampelisca* populations recolonized the community and became again the dominant species suggests that there was an evolutionary strategy related to population levels (Dauvin *et al.*, 1993). There were very limited changes of abiotic factors in the Bay of Morlaix. This area is influenced by Atlantic conditions occurring at the entrance of the English Channel (Dauvin, 1984). The abiotic and biotic factors affecting the community were conducive to the establishment and persistence of stable dominant *Ampelisca* populations in time. Nevertheless, Thorson (1957) suggested that the *Ampelisca* communities are not stable, because of the dense amphipod populations added to the characteristic fauna of soft-bottom communities. The 'climax' concept seems to be applicable in the case of the fine sand Pierre Noire community in such a stable environment. The soft-bottom *Abra alba* community of the Bay of Morlaix can only support a maximal carrying capacity of around 10–11 g m^{-2} (mean annual biomass in decalcified dry weight). The mixed trophic group depending on suspended and sediment surface matter was dominant and formed an important part of the total biomass and production. This suggests a high bentho-pelagic coupling, with benthic energy sources originating from the water column, stable through time. The important role of the carnivores in terms of biomass and production suggested an important link between this group and other macrofaunal and meiofaunal groups (Chardy and Dauvin, 1992). The deficit in production during 11 annual cycles may have affected the fish biomass of this community. Although the affected area covers a limited surface, it is an important zone of the Bay of Morlaix, and this incident could have had a significant consequence on fish catches for the north Brittany coast over more than 10 yr after the Amoco Cadiz event.

This study was carried out by the Programme National d'Océanographie Côtière (PNOC) IFREMER/INSU topic 'long-term series'. The data were obtained in the framework of the Programme GDR Manche, CNRS supported by IFREMER (RNO) contracts. The authors thank the captain and crew of 'NO Mysis', and the staff of the Marine Biological Station of Roscoff for field work. I gratefully acknowledge K. Ghertsos for checking the English.

Arntz, W. E. (1980) Predation by demersal fish and its impact on the dynamics of macrobenthos. In *Marine Benthic Dynamics*, ed. K. R. Tenore, pp. 121–129. University South Carolina Press, Columbia.

Arntz, W. E. and Brunswig, D. (1975) Studies on structure and dynamics of macrobenthos in the Western Baltic carried by the joint research programme 'interaction sea-sea bottom' (SBF95-Kiel). In *Proceedings of the 10th European Marine Biology Symposium*, ed. G. Persoone and E. Jaspers, pp. 17–42. Universa Press, Wetteren.

Cabioch, L., Dauvin, J. C. and Gentil, F. (1978) Preliminary observations on pollution of the sea bed and disturbance of sub-littoral communities in northern Brittany by oil from the 'Amoco Cadiz'. *Mar. Pollut. Bull.* **9**, 303–307.

Chardy, P. and Dauvin, J. C. (1992) Carbon flows in a subtidal fine sand community from the western English Channel: a simulation analysis. *Mar. Ecol. Prog. Ser.* **81**, 147–161.

Dauvin, J. C. (1984) Dynamique d'écosystèmes macrobenthiques des fonds sédimentaires de la baie de Morlaix et leur perturbation par les hydrocarbures de l'Amoco Cadiz. Thèse de doctorat Etat, ès sciences, Université de Paris VI, 456 pp+annexes 192 pp.

Dauvin, J. C. (1987) Evolution à long terme (1978–1986) des populations d'Amphipodes des sables fins de la Pierre Noire (Baie de Morlaix, Manche Occidentale) après la catastrophe de l'Amoco Cadiz. *Mar. Environ. Res.* **21**, 247–273.

Dauvin, J. C. (1988) Rôle du macrobenthos dans l'alimentation des Poissons démersaux vivant sur les fonds de sédiments fins de la Manche occidentale. *Cah. Biol. Mar.* **29**, 445–467.

Dauvin, J. C. (1991) Effets à long terme de la pollution de l'Amoco Cadiz sur la production de deux peuplements subtidaux de sédiments fins de la Baie de Morlaix (Manche Occidentale). In *Estuaries and Coasts: Spatial and Temporal Intercomparisons, Proceedings of the ECSA 19 Symposium*, eds. M. Elliot and J. P. Ducrotoy, pp. 349–358. Olsen and Olsen, Fredensborg, Denmark.

Dauvin, J. C., Bellan-Santini, D. and Bellan, G. (1993) Les genres *Ophelia* et *Ampelisca* de la région de Roscoff: exemples d'allotopie et de syntopie dans les communautés marines de substrat meuble. *Cah. Biol. Mar.* **34**, 1–15.

Holloway, M. and Horgan, J. (1991) Les marées noires. *Pour la Science* **170**, 78–88.

Laubier, L. (1991) Les marées noires. Conséquences à long terme. *La Recherche* **22**, 814–823.

Smith, J. E. (1968) *'Torrey Canyon' Pollution and Marine Life*. Cambridge University Press, 196 pp.

Thorson, G. (1957) Bottom communities (sublittoral or shallow shelf). *Mem. Geol. Soc. Am.* **67**, 461–534.